

// LEARNING PATH

Cybersecurity Roadmap

Foundation to Core Offensive Security to Specializations, converging on the CISE expert program.

12 COURSES · **191 MODULES** · **4 TIERS** · **18 MONTHS (CISE)**



Every course, every module

01 Foundation

5 courses

1. Enterprise Windows Infrastructure Security

BEGINNER TO ADVANCED · 14 MODULES

- 1 Networking Fundamentals › 2 Lab Setup and Virtualization › 3 Windows Server Management
- 4 Dynamic Host Configuration Protocol (DHCP) › 5 Domain Name System (DNS)
- 6 Windows Operating System Administration › 7 Active Directory Domain Services (AD DS)
- 8 Group Policy Objects (GPO) › 9 File Services & Distributed File System (DFS) › 10 Web Server (IIS)
- 11 FTP Server Administration › 12 Proxy Server Administration
- 13 Remote Access and VPN Configuration › 14 Windows Server Backup, Restore & Recovery

2. Linux Administration & Server Hardening

BEGINNER TO ADVANCED · 18 MODULES

- 1 Introduction to Linux › 2 Linux Basic Commands › 3 Text Editors
- 4 String Processing & Finding Files › 5 Users, Groups & Permissions › 6 Package Management
- 7 File System & Disk Management › 8 Network Configuration & Controlling Services
- 9 Security, Process Management & Monitoring › 10 Dynamic Host Configuration Protocol (DHCP)
- 11 Domain Name System (DNS) › 12 Apache Web Server › 13 FTP Server (VSFTPD)
- 14 Samba/SMB/CIFS Server › 15 NFS Server › 16 Telnet & Remote Desktop Server
- 17 Proxy Server › 18 TFTP & PXE Boot Server

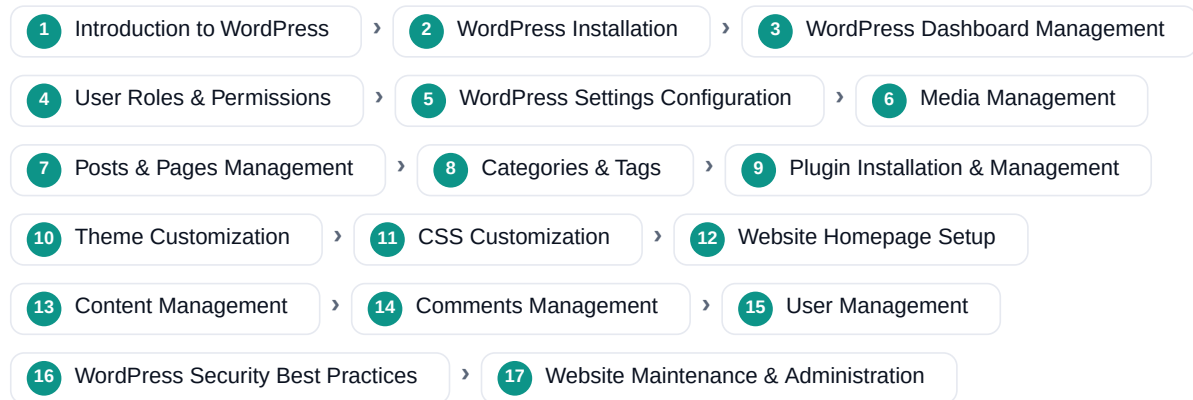
3. Secure PHP Development

BEGINNER TO INTERMEDIATE · 19 MODULES

- 1 Introduction to PHP › 2 First Steps in PHP › 3 Variables & Data Types
- 4 Arrays & Associative Arrays › 5 Control Structures › 6 Loops & Iterations
- 7 User-Defined Functions › 8 Scope & Global Variables › 9 Debugging & Error Handling
- 10 Building Dynamic Web Pages › 11 Working with Forms & Validation › 12 Cookies & Sessions
- 13 MySQL Fundamentals › 14 CRUD Operations › 15 PHP & MySQL Integration
- 16 SQL Injection Prevention › 17 Prepared Statements › 18 File & Directory Handling
- 19 Secure PHP Development Practices

4. Secure WordPress Administration

BEGINNER · 17 MODULES



5. Python for Security Professionals

BEGINNER TO ADVANCED · 14 MODULES



Certified Ethical Hacking & Penetration Testing

BEGINNER TO ADVANCED · 16 MODULES

- 1 Kali Linux Fundamentals › 2 Introduction to Penetration Testing & Ethical Hacking
- 3 Footprinting & Reconnaissance › 4 Scanning Networks › 5 Proxies, VPNs & Tor
- 6 Tunneling Techniques › 7 Enumeration › 8 System Hacking & Malware Threats
- 9 Privilege Escalation › 10 Buffer Overflow › 11 Advanced Exploitation & Metasploit Framework
- 12 Evading IDS, Firewalls & Honeypots › 13 Sniffing & MITM Attacks › 14 Social Engineering
- 15 Denial of Service (DoS/DDoS) › 16 Cryptography

1. Wireless Security & WiFi Pentesting

ADVANCED · 14 MODULES

- 1 Introduction to Wireless Networks › 2 Wireless Encryption & Authentication
- 3 Wireless Network Cards in Linux › 4 Wireless Security Measures & Bypass
- 5 Wireless Reconnaissance & Traffic Analysis › 6 Wireless Denial-of-Service Attacks
- 7 Wireless MITM & Rogue Access Points › 8 WEP Cracking Techniques
- 9 Chop-Chop & Packet Replay Attacks › 10 Caffe Latte Attack › 11 WPA/WPA2 Cracking
- 12 Cowpatty & Hash Table Attacks › 13 Advanced WPA/TKIP Attacks
- 14 Enterprise Wireless Security & Reporting

2. Active Directory Security & Enterprise Attacks

ADVANCED · 13 MODULES

- 1 LDAP Enumeration › 2 Kerberos Authentication › 3 Kerberoasting Attacks
- 4 Pass-the-Ticket (PtT) › 5 AS-REP Roasting › 6 Manual Enumeration
- 7 Password Attacks & Credential Dumping › 8 Pass-the-Hash (PtH)
- 9 LLMNR Poisoning & SMB Relay Attacks › 10 Active Directory Offensive Security Tools
- 11 Active Directory Privilege Escalation › 12 Enterprise Lateral Movement & Persistence
- 13 Active Directory Hardening & Reporting

3. Advanced Web Application Security

ADVANCED · 17 MODULES

- 1 Web Server Concepts › 2 Web Server Hardening & Security
- 3 Web Server Enumeration & Misconfigurations › 4 Web Application Security Tools
- 5 Web Application Concepts & Architecture › 6 Web Application Security Fundamentals
- 7 Web Application Testing Methodology › 8 Injection Attacks › 9 Authentication & Session Attacks
- 10 Cross-Site Scripting (XSS) › 11 Security Misconfiguration › 12 Sensitive Data Exposure
- 13 File Inclusion & Path Traversal › 14 SSRF & XXE Attacks › 15 Access Control & IDOR
- 16 Advanced Web Exploitation Techniques › 17 Additional Web Attack Vectors

4. API Security & Advanced Exploitation

INTERMEDIATE TO ADVANCED · 22 MODULES

- 1 Introduction to API Penetration Testing › 2 Lab Setup & Testing Environment › 3 SOAP API Security
- 4 REST API Security › 5 GraphQL API Security › 6 API Reconnaissance
- 7 Endpoint Analysis & Discovery › 8 JWT Vulnerabilities & Exploits › 9 API Injection Attacks
- 10 Broken Object Level Authorization (BOLA) › 11 Broken Authentication
- 12 Exploiting API Authorization › 13 Broken Object Property Level Authorization (BOPLA)
- 14 Broken Function Level Authorization (BFLA) › 15 Rate Limiting & Resource Consumption
- 16 Business Logic & Sensitive Flows › 17 Server-Side Request Forgery (SSRF)
- 18 Security Misconfiguration › 19 Improper Inventory & Asset Management
- 20 Unsafe Consumption of APIs › 21 GraphQL-Specific Attacks › 22 Reporting & Remediation

5. Mobile Application Penetration Testing

ADVANCED · 14 MODULES

- 1 Mobile Security Landscape & Threat Modeling › 2 Android Platform Internals
- 3 iOS Platform Internals › 4 Lab Setup: Devices, Emulators & Tooling › 5 Android Static Analysis
- 6 iOS Static Analysis › 7 Dynamic Analysis & Runtime Instrumentation
- 8 Root & Jailbreak Detection Bypass › 9 Certificate Pinning Bypass › 10 Insecure Data Storage
- 11 IPC & Component Vulnerabilities › 12 Runtime Manipulation & Tampering
- 13 Mobile Backend API Testing › 14 Reporting & OWASP MASVS Verification

6. AI/ML Penetration Testing

ADVANCED · 13 MODULES

- 1 AI/ML Security Landscape & Threat Modeling › 2 MITRE ATLAS & NIST AI RMF
- 3 LLM Prompt Injection › 4 Jailbreaks & Guardrail Bypass › 5 RAG Pipeline Attacks
- 6 MCP & Agentic System Abuse › 7 Adversarial Examples for ML Classifiers
- 8 Training-Data Poisoning & Backdoors › 9 Model Extraction & Model Inversion
- 10 ML Supply Chain Attacks › 11 LLM Application Security Testing › 12 Tooling Deep Dive
- 13 Reporting, Governance & AI Red Team Operations

// CAPSTONE · CISE

Certified Information Security Expert

The complete program that unifies every track into one career-defining qualification — all 12 courses · 191 modules.